# Big Mountain Resort Pricing Recommendation Report

*Guided Capstone Step 6 - simple summary with recommendations*

**Main recommendation:** Big Mountain looks underpriced. I recommend a careful ticket increase from $81 to about $86-$90 first, then review demand before moving closer to the model-supported level of about $96.

## Problem statement

The goal of this project was to recommend an adult weekend ticket price for Big Mountain Resort. I used data from other ski resorts to estimate what price Big Mountain's facilities can support in the market.

## Data wrangling, EDA, and preprocessing

The work started with 330 resorts in the raw data and ended with 277 usable resorts after cleaning. I removed weak or missing data, merged state-level summary data, and created helpful ratio features such as chairs per run. The exploratory analysis showed that ticket price generally rises with stronger resort features, especially vertical drop, runs, chair lifts, skiable terrain, fast quads, and snow making. For modeling, I removed text fields like resort name, state, and region, then used median imputation for missing values.

## Algorithms used and model evaluation

I compared three approaches: a simple average-price baseline, linear regression, and random forest regression. I used mean absolute error (MAE) as the main evaluation metric because it is easy to understand in dollars. The baseline missed by about $19 on average, so it was too weak. Linear regression improved a lot, but random forest gave the best and most stable results, so it was chosen as the final model.

| Model | CV MAE | Test MAE | Comment |
|---|---|---|---|
| Average baseline | - | $19.14 | Too weak |
| Linear regression | $10.50 | $11.79 | Good but less accurate |
| Random forest | $9.64 | $9.54 | Best final model |

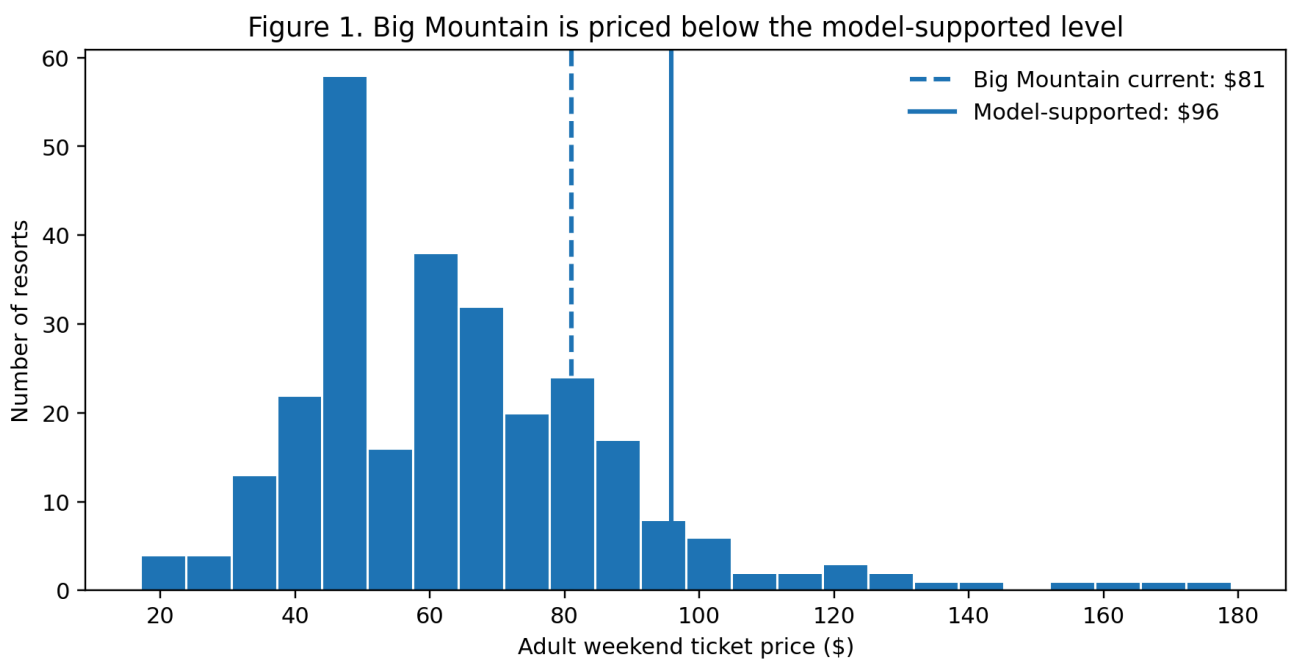


*Figure 1. Big Mountain currently charges $81. The model suggests the resort could support about $95.87.*

## Winning model, scenario modeling, and recommendation

After the final model was trained on all resorts except Big Mountain, it predicted a supported ticket price of about $95.87 for Big Mountain. The model's cross-validation MAE was about $10.39, so I would not jump straight to $96. A

better business decision is a step-by-step test: raise the adult weekend price first to about $86-$90, track ticket sales, guest feedback, and revenue, and then consider moving toward $92-$96 if demand stays strong.

The scenario work gives two more lessons. First, Big Mountain already ranks very high on key features, which supports a higher price. Second, scenario 2 (adding one run, 150 feet of vertical drop, and one extra chair lift) adds about $1.99 of supported price per ticket, or about $3.47 million in seasonal revenue if 350,000 visitors buy five tickets each. However, I would only study this option further after reviewing the lift's real operating and capital costs. Scenarios 3 and 4 did not add meaningful extra value. Closing one run had no effect, but closing 3 to 10 runs reduced the supported ticket price by about $0.67 to $1.81, so any run-closure test should be small and carefully monitored.

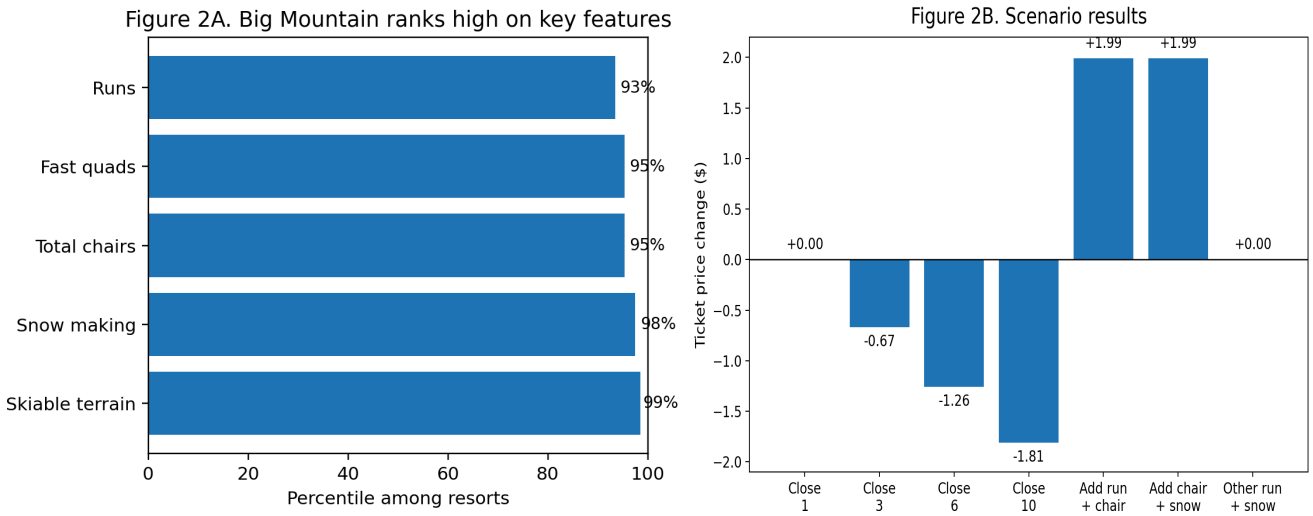


*Figure 2. Big Mountain ranks in the top part of the market on several key features, and most scenario changes have only small price effects.*

## Conclusion and future work

My final recommendation is to raise Big Mountain's adult weekend ticket price gradually instead of making one large jump. The data suggests the resort is underpriced relative to similar resorts, but the model still has error and cannot capture every business factor. Future work should add cost data, local competition, discount strategy, lodging capacity, and customer demand information. That would help move from a price recommendation to a stronger profit recommendation.